



AI-Assisted Disaster Mapping with NASA OPERA Data in GeoLibre

Qiusheng Wu
University of Tennessee, Knoxville
<https://qishub.org>



What is GeoLibre



A free and open-source, lightweight, **cloud-native GIS platform** for visualizing, exploring, and analyzing geospatial data.

GeoLibre runs everywhere you do, in the web browser, on the desktop, on mobile, and inside Jupyter notebooks, all while keeping your data local and private.

Website: <https://geolibre.app>

Web App: <https://web.geolibre.app>

GitHub: <https://github.com/opengeos/GeoLibre>



No account

Open the page and it is ready.
No sign-up, no quota, no trial period.

No server

No GeoServer, no database,
no tile cache, and therefore
nothing to operate.

No upload

Computation happens locally
and data stays in the browser
by default, which makes
privacy and compliance
straightforward.

No install

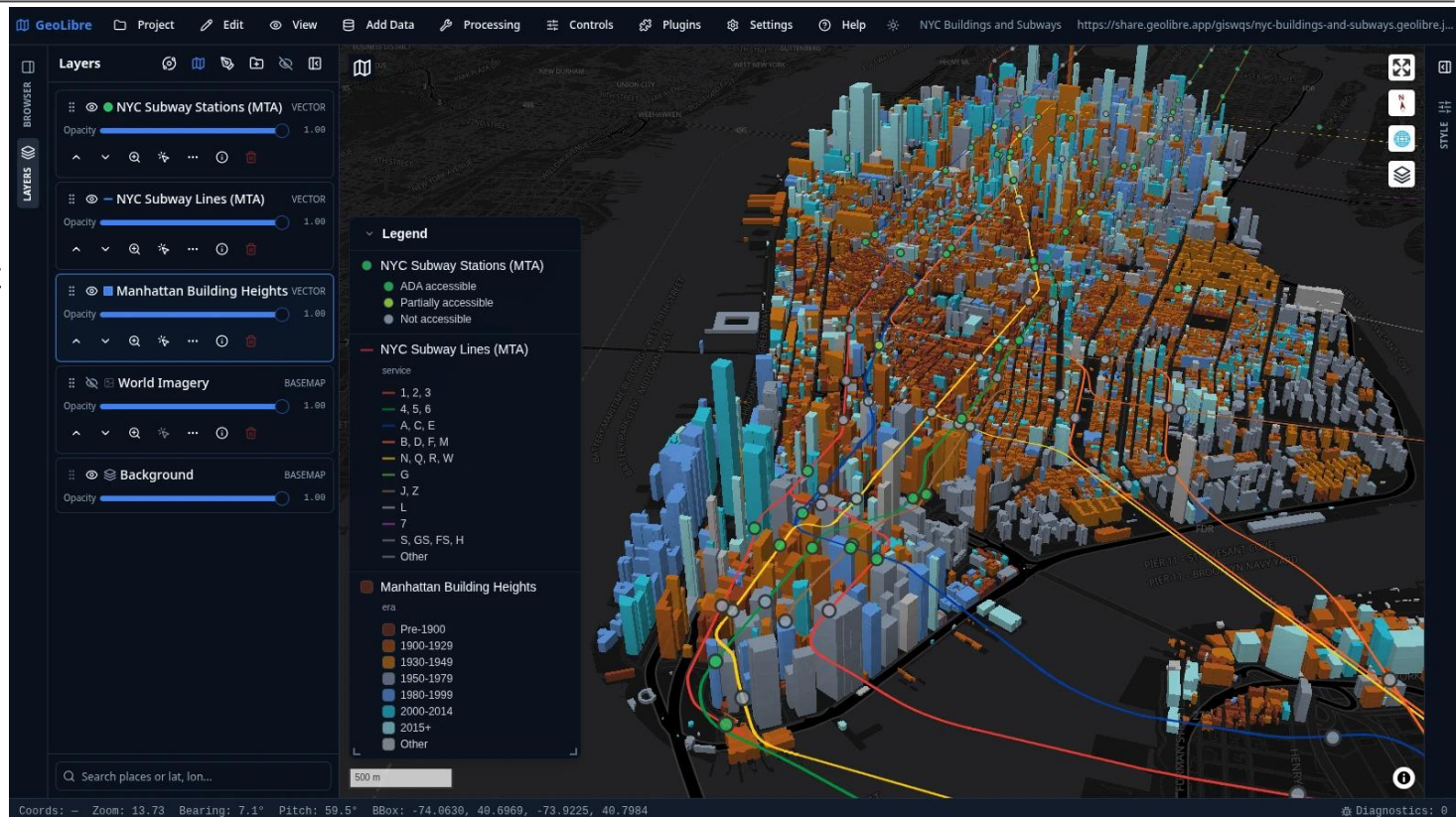
The web build installs nothing.
Add the desktop app when
you want more, and the
interface is identical.

Geospatial Data Visualization



Supported formats:

- GeoJSON
- Shapefile
- GPKG
- GeoParquet
- FlatGeobuf
- COG
- PMTiles
- COPC
- 3D Tiles
- Zarr
- NetCDF
- XYZ
- WMS
- WFS
- ...

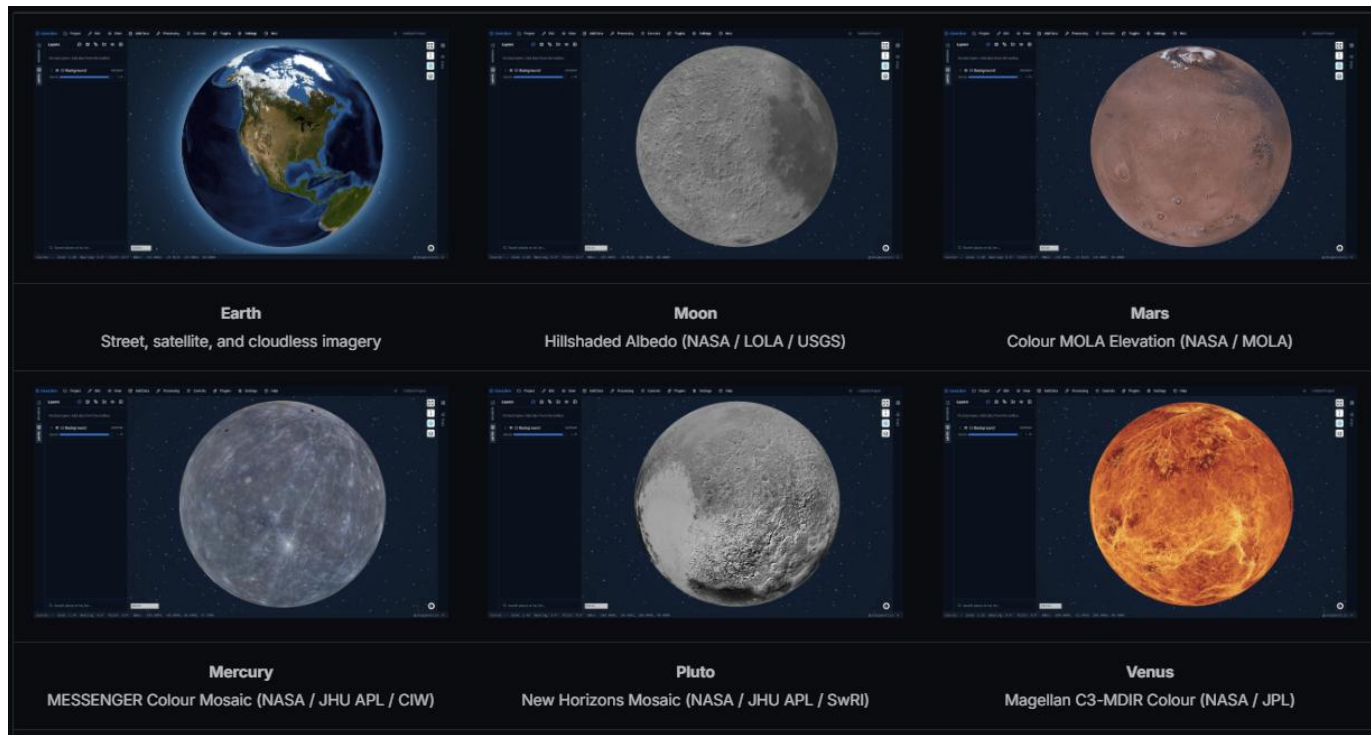


Not Limited to Earth



Planetary basemaps:

- Earth
- Moon
- Mars
- Mercury
- Venus
- Europa
- Ganymede
- Callisto
- Titan
- Pluto
- Charon



1,000+ Geoprocessing Tools



Toolbox:

- Vector
- Raster
- Conversion
- Hydrology
- LiDAR
- Network
- Projection
- Remote Sensing
- Terrain

Model Builder

AI Assistant

The screenshot displays the GeoLibre software interface, which is a web-based geospatial processing platform. The interface is divided into several panels:

- Layers Panel:** Located on the left, it shows a list of layers including "Extracted Stream Network" (VECTOR), "dem.tif" (COG), and "Background" (BACKGROUND). Each layer has an opacity slider and a "1.00" value.
- Model Builder Panel:** The central panel, titled "Extract Stream Network from DEM", contains a workflow diagram. The workflow starts with an "INPUT dem.tif" box, followed by a "WHITEBOX Fill Depressions Wang An..." box, then a "WHITEBOX D8 Pointer" box, and a "WHITEBOX D8 Flow Accum" box. These are connected to a "WHITEBOX Raster Streams To Vector" box, which finally outputs to an "OUTPUT Extracted Stream Network" box. A "CONVERSION" list on the left of this panel includes tools like "Add Point Coordinates To Table", "Clean Vector", "Convert Nodata To Zero", etc.
- Output Panel:** Located on the right, it shows the "Result name" as "Extracted Stream Network" and a "Saved models" section with a "Load a saved model." button.
- Map View:** At the bottom, a map shows a terrain visualization with a stream network overlaid. A scale bar indicates "5 km".
- Console/Log Panel:** At the bottom, it displays the execution log for the "Running Fill Depressions Wang And Liu..." process, showing file paths and conversion steps.

The top menu bar includes options like Project, Edit, View, Add Data, Processing, Controls, Plugins, Settings, and Help. The bottom status bar shows coordinates, zoom level, bearing, pitch, and bounding box.

Explore OPERA Data in GeoLibre



A cloud-native GIS platform for visualizing, exploring, analyzing, and sharing OPERA data.

GeoLibre -> **Plugins** -> **Web Services** -> **Activate Plugin (NASA Earthdata / Earthdata GIS)**

GeoLibre -> **Settings** -> **Manage Plugins** -> **Install NASA OPERA Plugin**

PLUGIN	SERVES	GRANULARITY	BEST FOR
NASA Earthdata	GIBS pre-rendered global imagery tiles	Daily global mosaic	The fast look - context and backdrop
Earthdata GIS	ArcGIS Image / Map / Feature services	Curated service, incl. NASA Disasters event groups	Authoritative event layers - FEDS perimeters, burn severity
NASA OPERA	CMR granule search + titiler-cmr COG tiles	Individual granule, individual band	The analysis itself - specific scenes, specific dates

NASA Earthdata Plugin



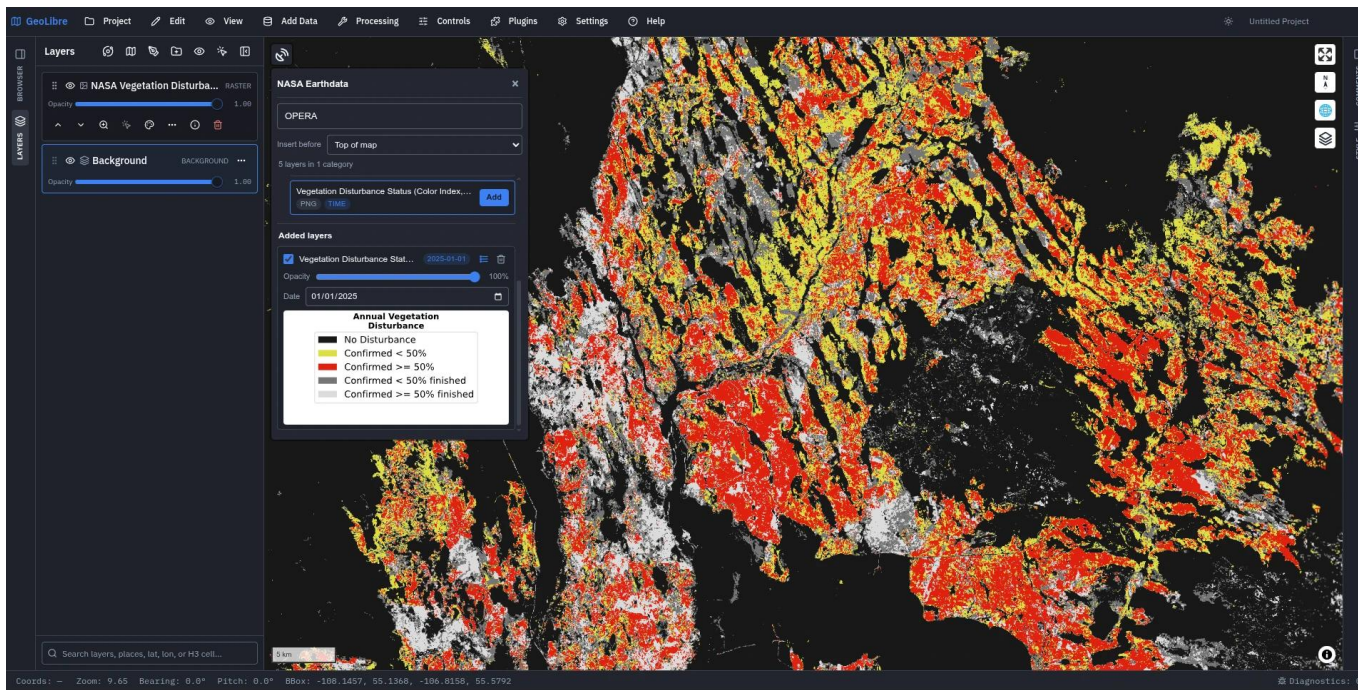
NASA **GIBS** - the Global Imagery Browse Services tile catalog, browsable from a map panel.

Strength:

Instant. No search, no granule, no auth - a global backdrop in one click, ideal for situational context around an event.

Limit:

Pre-rendered pixels. You cannot restretch a band, query a value, or run statistics - it is a picture, not a measurement.



Earthdata GIS Plugin



Search gis.earthdata.nasa.gov and add EOSDIS ArcGIS services straight to the map.

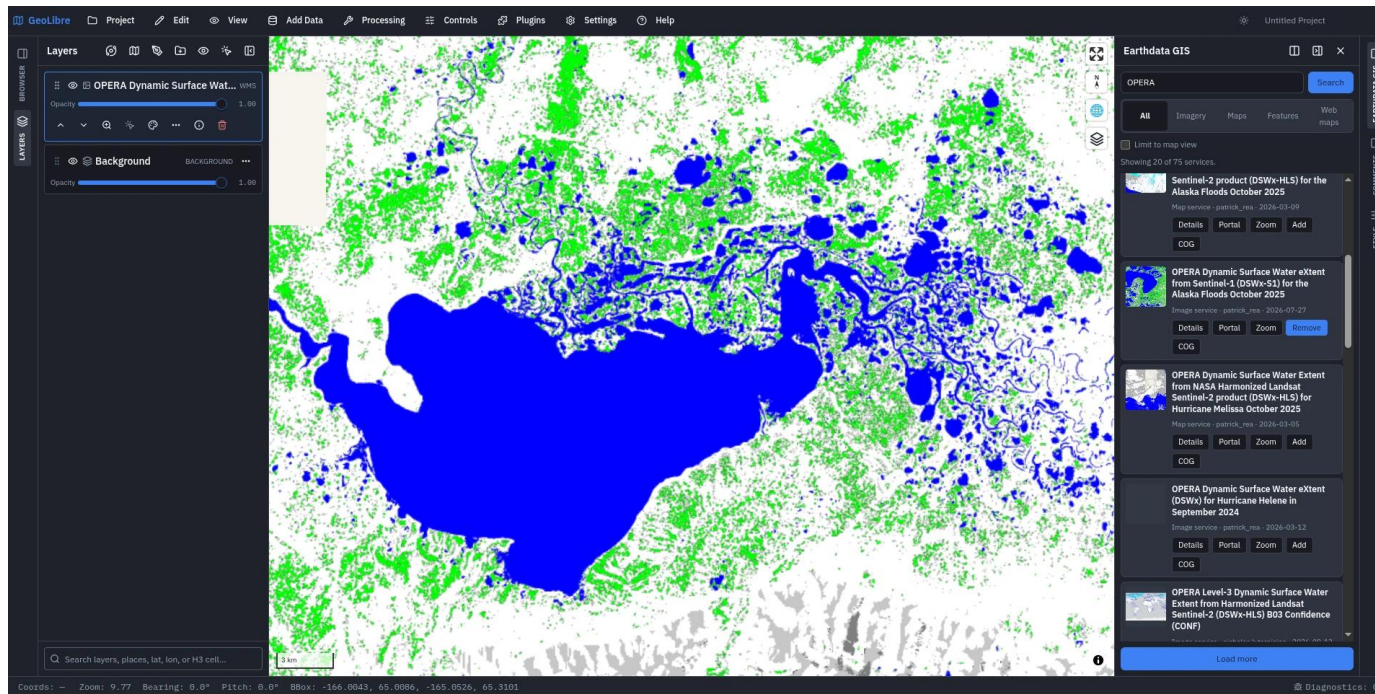
Strength:

Curated event-based data layers:

- Inundation extent
- Burn severity
- FEDS fire perimeter
- ...

Limit:

Some imagery layers are only available as WMS, not COG.



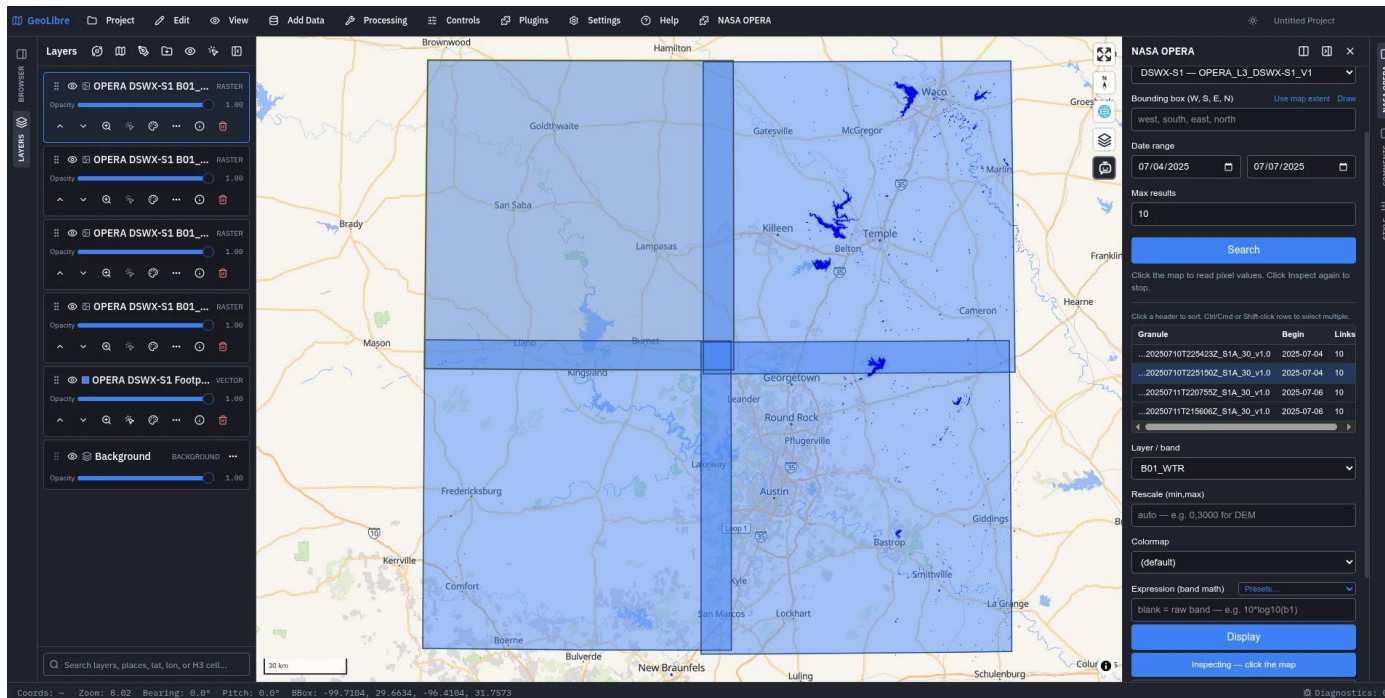
NASA OPERA Plugin



Granule-level search and display for eight OPERA collections.

Functionalities:

- Search
- Select
- Display
- Rescale
- Colormap
- Statistics
- Download



OPERA GeoAgent



A chat control on the map that can inspect and operate the live map.

Data Sources:

- Planetary Computer
- NASA OPERA

AI Providers:

- Anthropic
- OpenAI
- Gemini
- Amazon Bedrock
- Ollama
- OpenAI-compatible

The screenshot displays the OPERA GeoAgent interface within a web browser. The main map shows a satellite view of Las Vegas, Nevada, with a yellow bounding box highlighting a specific area. A chat window titled "OPERA GeoAgent" is open, showing a prompt: "Display a cloud-free Sentinel-2 image over Las Vegas for August 2026". The chat window also displays the provider and model information, the product (OPERA_L3_DSWX-HLS_V1), the band (B01_WTR surface water), and the granules displayed (3). The interface includes a layers panel on the left with various data sources like "OPERA DSXW-HLS B01_WTR" and "Sentinel-2 true color". The right panel shows the "NASA OPERA" settings, including the product, bounding box, date range, and a table of granules. The bottom status bar shows coordinates and map controls.

OPERA GeoAgent

Provider & model

Displayed OPERA DSXW-HLS imagery for the corresponding date, August 2, 2026, over Las Vegas.

Product: OPERA_L3_DSWX-HLS_V1

Band: B01_WTR surface water

Granules displayed: 3

Layer IDs:

opera-cog-OPERA-L3-DSWX-HLS-T11SQV-20260802T181921Z-20260804T030716Z-S2C-30-v1-1-B01-WTR

opera-cog-OPERA-L3-DSWX-HLS-T11SPV-20260802T181921Z-20260804T030723Z-S2C-30-v1-1-B01-WTR

opera-cog-OPERA-L3-DSWX-HLS-T11SPA-20260802T181921Z-20260804T044528Z-S2C-30-v1-1-B01-WTR

Draw AOI Clear AOI No precise AOI

Prompt

Display a cloud-free Sentinel-2 image over Las Vegas for August 2026

Send Cancel Copy Markdown Clear

NASA OPERA

Flood benchmark (QAEAD) Clear

Import benchmark ...

No benchmark locked.

Product

DSWX-HLS — OPERA_L3_DSWX-HLS_V1

Bounding box (W, S, E, N) Use map extent Draw

-115.45, 35.95, -114.85, 36.45

Date range

08/02/2026 08/02/2026

Max results

10

Search

Displayed 3 granules(s).

Click a header to sort. Click a row to select multiple.

Granule	Begin	Links
...20260804T044528Z_S2C_30_v1-1	2026-08-02	20
...20260804T030716Z_S2C_30_v1-1	2026-08-02	20
...20260804T030723Z_S2C_30_v1-1	2026-08-02	20

Layer / band

B01_WTR

Rescale (min,max)

auto — e.g. 0.3000 for DEM

Colormap

(default)

Expression (band math) Presets

blank = raw band — e.g. 10*log10(b1)

Diagnosics: 0

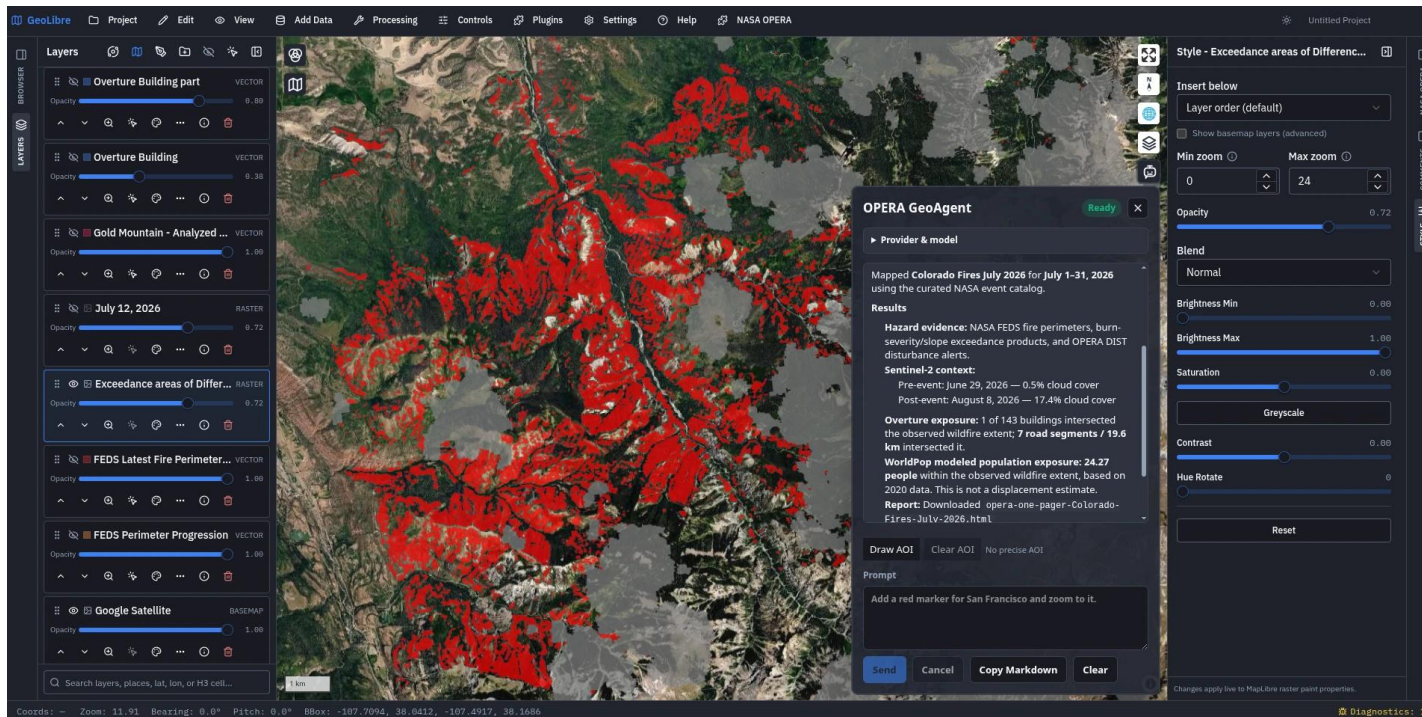
Disaster Mapping



Disaster mapping using plain language. One sentence: A place, a hazard, and a time window.

Example:

- Map Colorado fire in July 2026

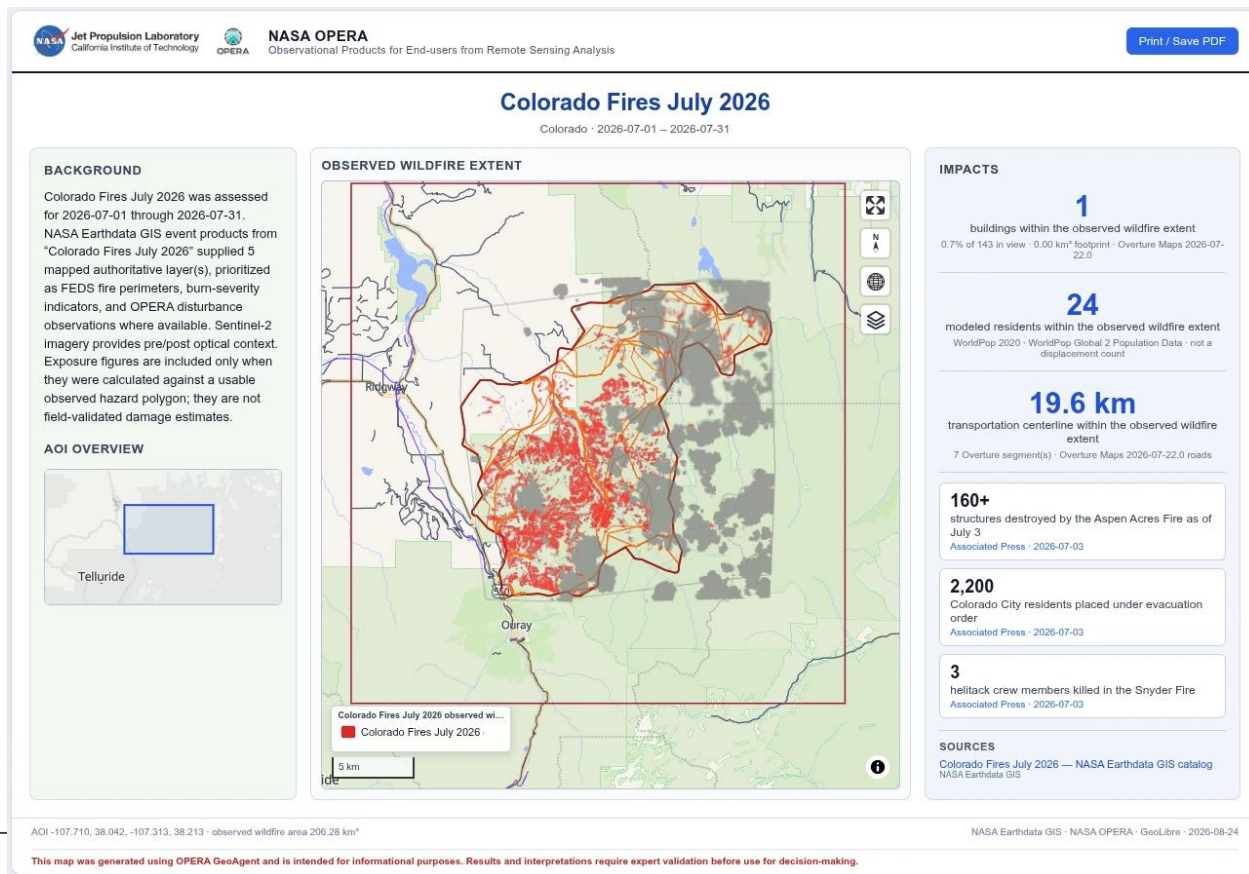


Self-contained HTML and PDF



Sections:

- Background
- AOI overview
- Observed hazard extent
- Impacts
- Sources
- Disclaimer





Live Demo

<https://opera-plugin.geolibre.app>





Q&A
